Contact: heesok@andrew.cmu.edu, jfeather@andrew.cmu.edu

Probing Invariances in Auditory Event Categorization using Model Metamers

Hee So Kim[1], Abigail L. Noyce[1], Malinda J. McPherson-McNato[2], Jenelle Feather[1]

1 - Carnegie Mellon University, Pittsburgh, PA, USA. 2 - Purdue University, West Lafayette, IN, USA.

Carnegie Mellon University

Introduction

Goal: Understand how the auditory system forms invariant representations of natural sounds

- Invariance (e.g., to pitch and timing) is key to robust auditory event categorization
- DNNs trained to recognize sound categories can model stages of auditory processing

Questions:

(1) Does adversarial training help align model and human invariances for audio event models?

(2) Are model invariances, as measured by model metamers, task-specific?

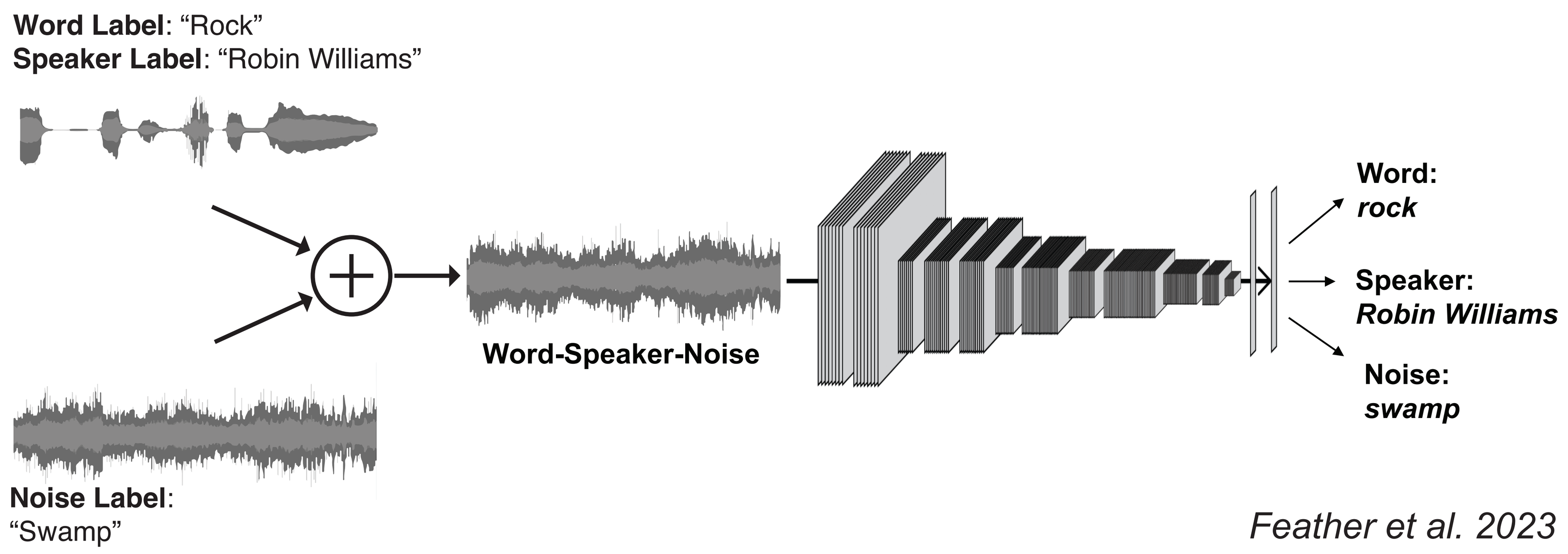
Approach:

Compare human and model responses to matched stimuli, both natural and "metameric" (model-equivalent) versions

Method: Model Selection

Tested multiple pretrained DNNs [1]:

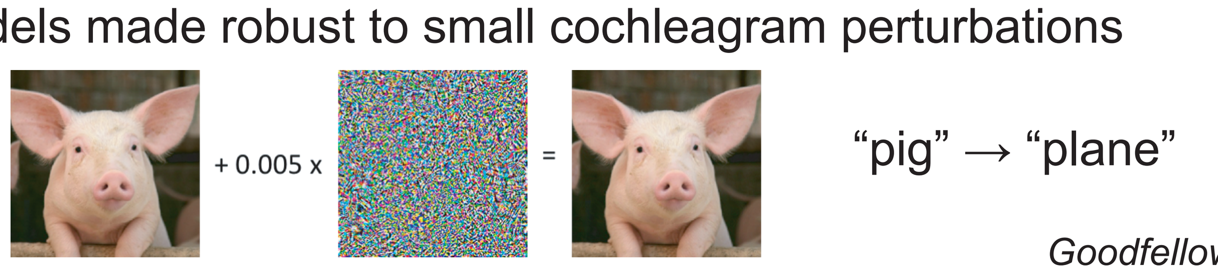
- **Word-AT:** Word recognition with adversarial training
- **Event-AT:** Auditory event recognition with adversarial training
- **Multi-AT:** Multi-task model with adversarial training
- **Multi-S:** Multi-task model with standard training



Feather et al. 2023

Adversarial training (AT) improves alignment to human perception [1,2]

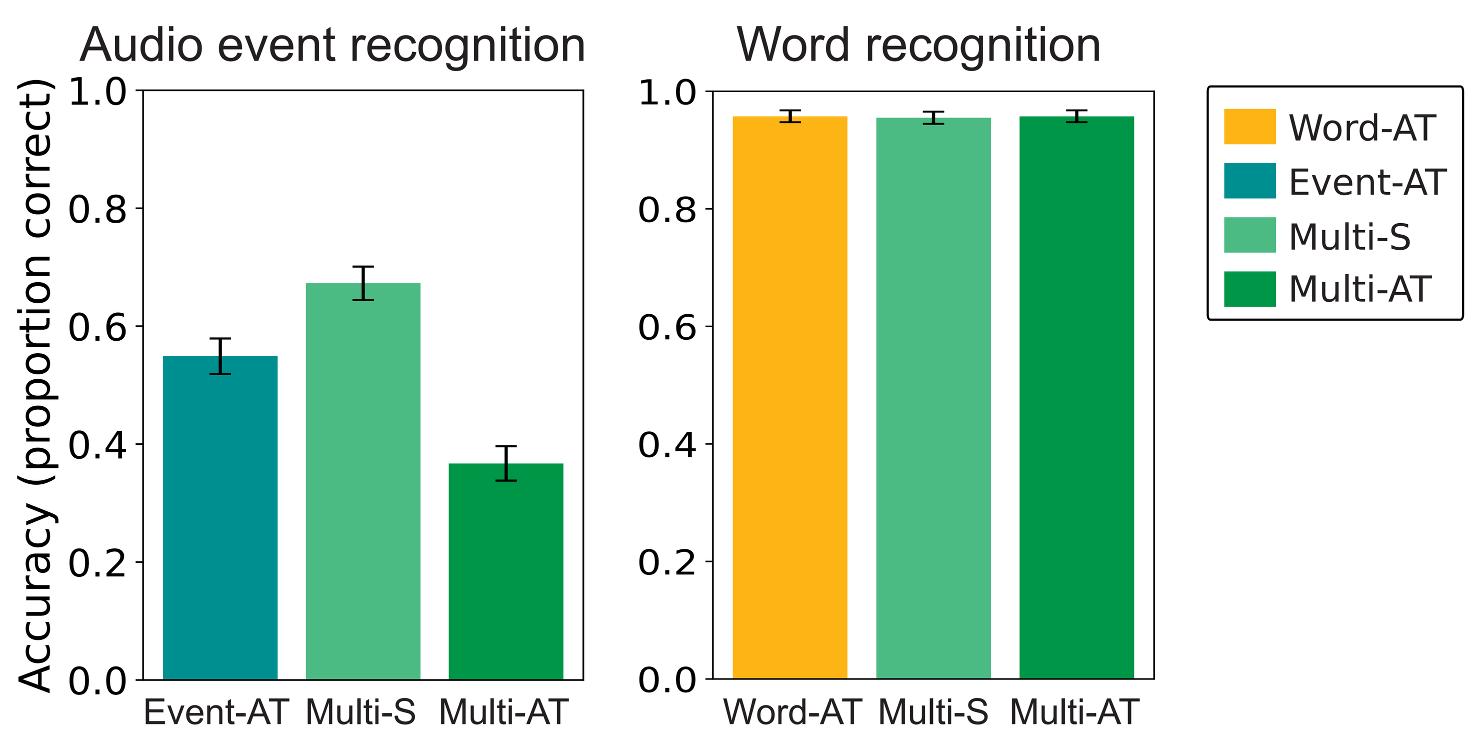
- AT models made robust to small cochleagram perturbations



Goodfellow et al. 2015

Models perform relatively well on the tasks they were trained for

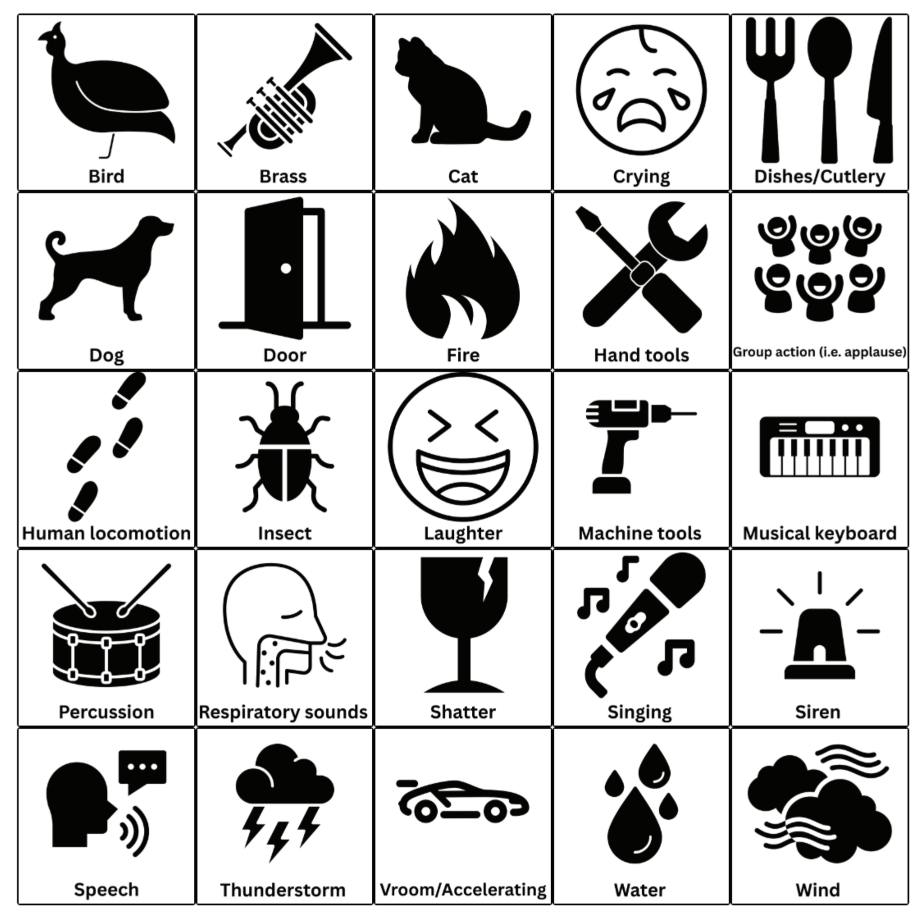
- Auditory event recognition: tested on 275 sounds from FSD50K dataset
- Word recognition: tested on 400 words from Wall Street Journal Corpus [1]

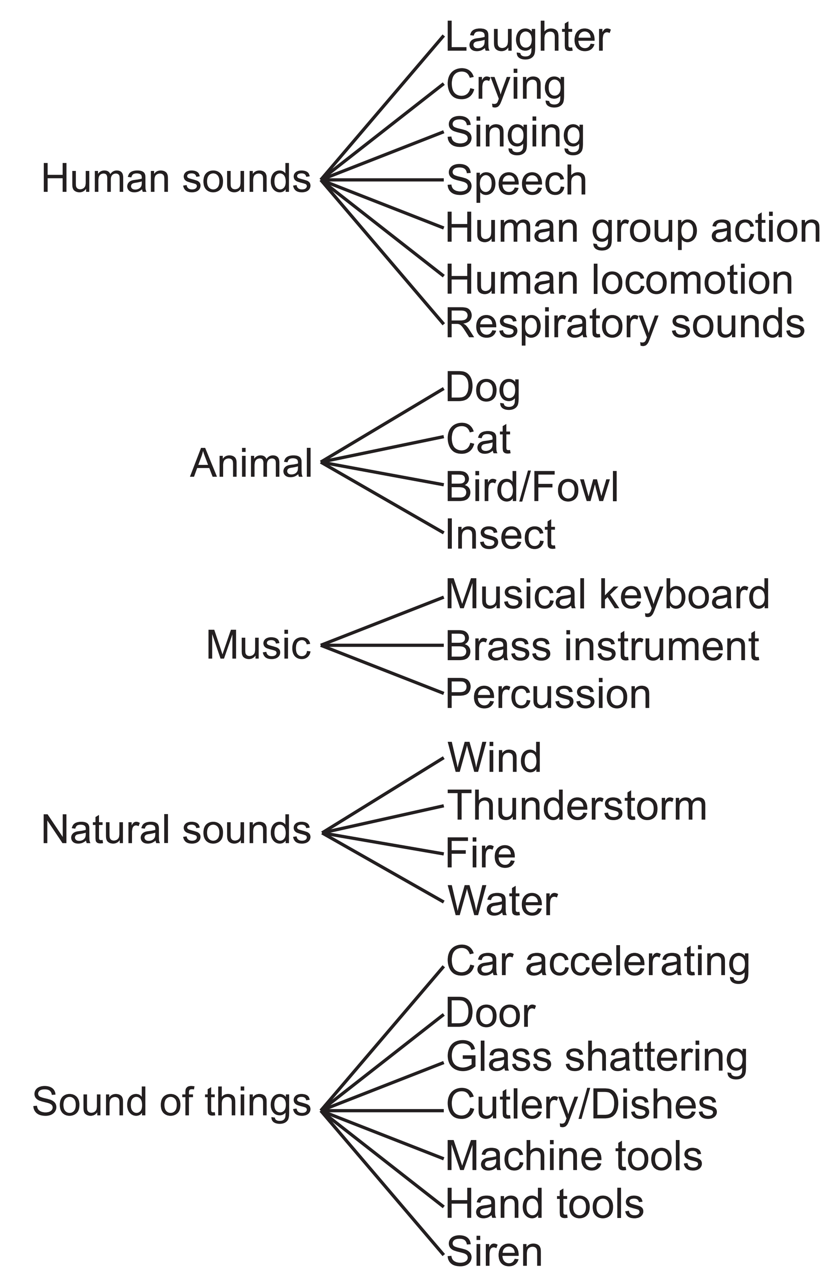


Method: Behavioral Task

- Developed a novel 25-way force choice sound classification task
- Selected 25 distinct categories, following structure of AudioSet ontology [3]
- Filtered sounds from FSD50K dataset [4] for 11 examples per category
- Used 2-second samples of sounds for experiments

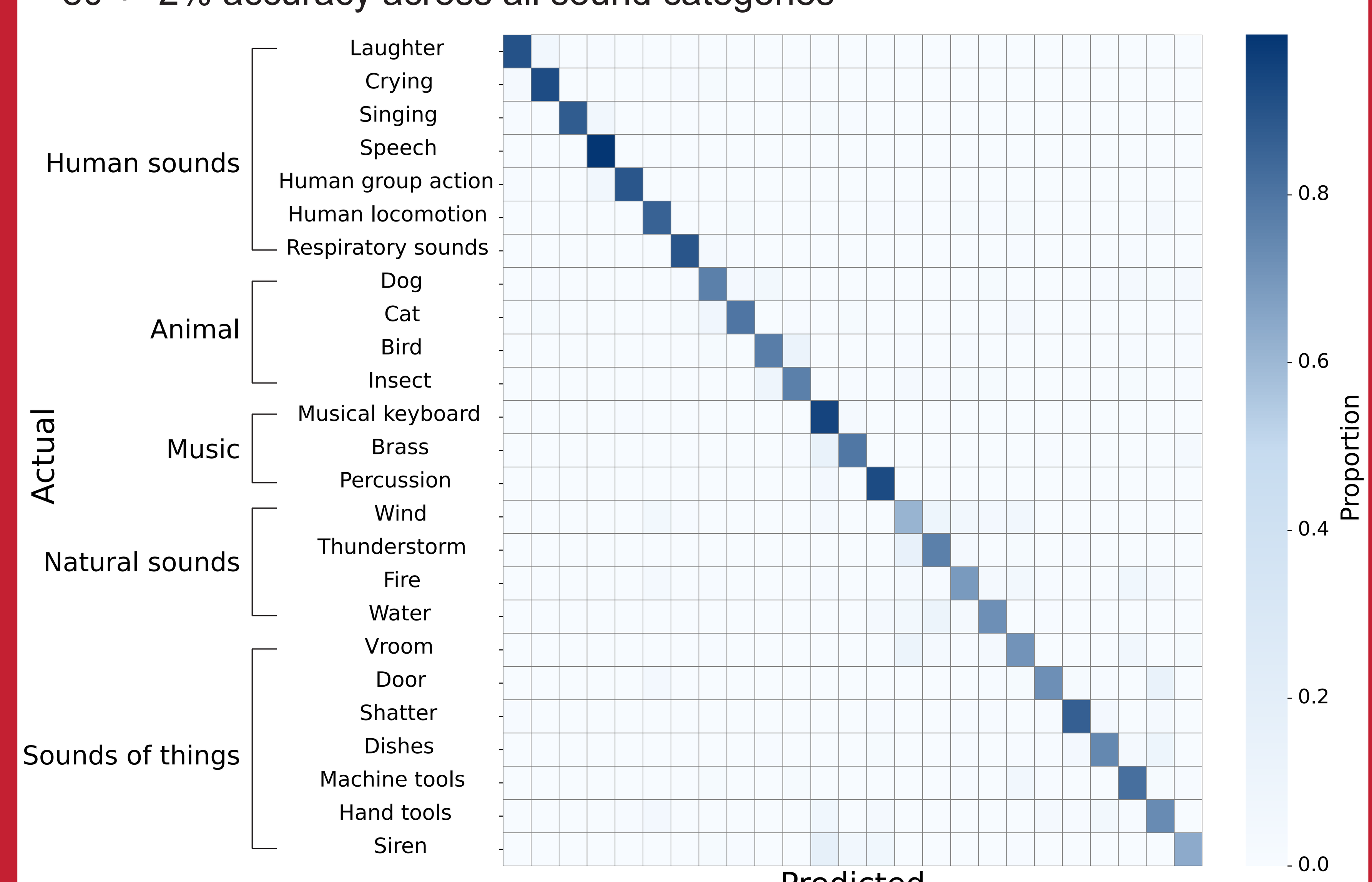
Choose the category corresponding to the sound





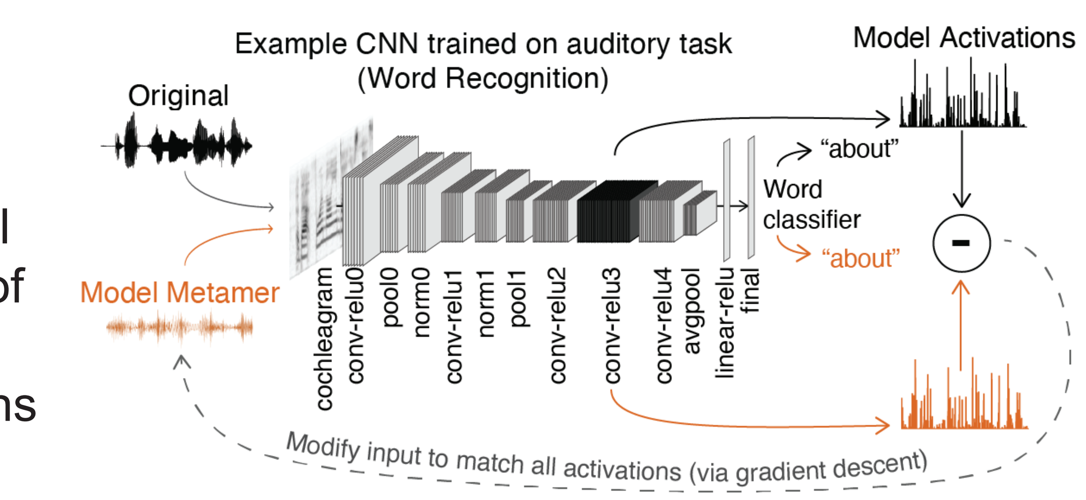
Humans classify natural sounds into the selected 25 categories

- 286 trials
  - 11 per category, plus 11 catch trials
- N=20 participants (collected on Prolific)
- 80 +- 2% accuracy across all sound categories



Method: Testing invariances with model metamers

- **Model Metamers**: Two distinct stimuli that produce the same response in a computational model
- Human recognition of model metamers tests whether humans share model invariances

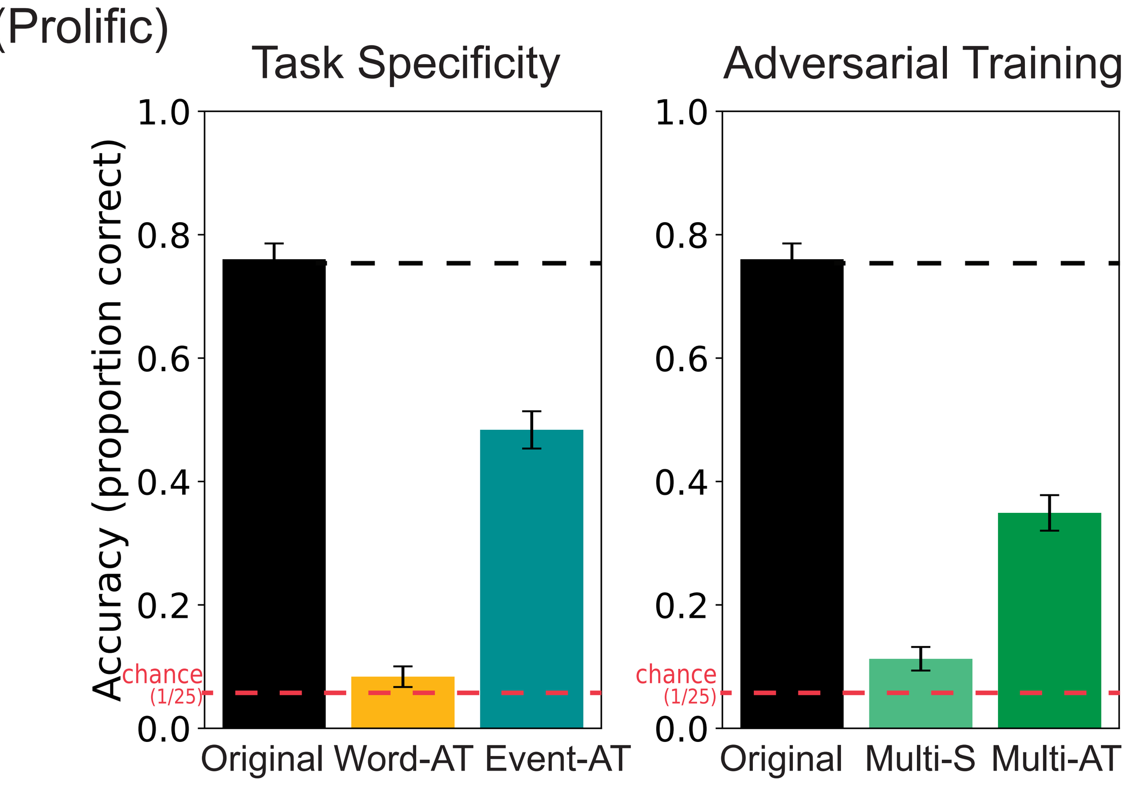


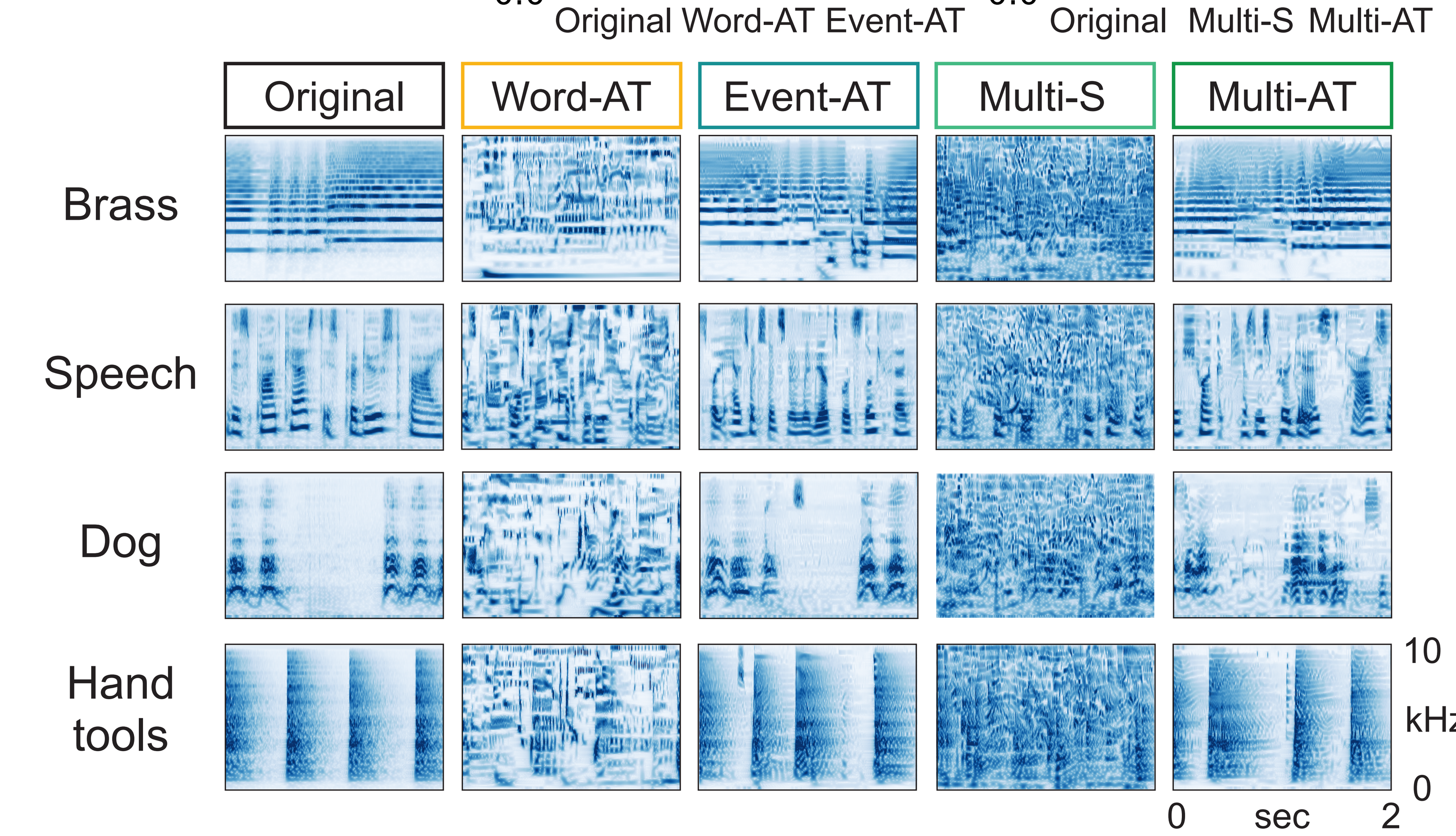
Feather et al. 2023

Model invariances are shaped by both adversarial training and task

- Adversarial training improves human recognition of model metamers
  - similar effect for word-trained models in [1]
- Model task shapes the invariance
  - Model metamers for audio events are more recognizable from the audio event trained model than the word-trained model
- n=4 participants (Prolific)

Model Metamer Sound Examples





Conclusion

- We developed and validated a novel paradigm to test if models share invariances with humans in audio event categorization
- Humans can reliably recognize model metamers only if the model was (1) trained on the task and (2) adversarially trained

References:
[1] Feather et al. 2023
[2] Goodfellow et al. 2015
[3] Gemmeke et al. 2017
[4] Fonseca et al. 2021

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